



**Indian Institute of Technology Guwahati**

**Department of Electronics and Electrical Engineering**

**Digital IC Design Laboratory**

Lab Report

**Course Code: EE518**

**Semester: 1**

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**Roll Number: 254102404**

## EXPERIMENT-05

### Aim:

To design and implement the layout of a Radix-4 Booth 8-bit multiplier using Cadence Virtuoso tool through DRC, LVS.

### Theory:

The Radix-4 Booth Multiplier is an efficient algorithm used for binary multiplication, designed to reduce the number of partial products and thereby speed up the multiplication process. It is an extension of the original Booth's algorithm (Radix-2) that groups the multiplier bits in overlapping sets of three, allowing the algorithm to handle two bits of the multiplier at a time.

### Working Principle

The multiplier bits are examined in overlapping triplets, where each triplet consists of two current bits and one preceding bit (initially zero for the least significant bits). Each triplet determines one partial product to be generated based on the following logic:

- The partial product can be 0,  $\pm 1$ , or  $\pm 2$  times the multiplicand.
- Negative values are handled using two's complement to simplify hardware design.
- By processing two bits at a time rather than one, the number of partial products is approximately halved compared to a Radix-2 multiplier.

### Key Steps in the Algorithm

1. Preprocessing: The multiplier's sign bit is extended if necessary, and a zero bit is appended to its least significant end.
2. Grouping: The multiplier bits are scanned in overlapping groups of three bits.
3. Partial Product Generation: Each group of three bits is used to select a partial product as multiples of the multiplicand according to a predefined encoding table.
4. Shifting: Each partial product is shifted left by a multiple of two bits, corresponding to the position of the processed group in the multiplier.

5. Summation: All partial products are summed to obtain the final product.

#### Advantages

- Significant reduction in the number of partial products leads to faster multiplication.
- Simplified hardware implementation due to fewer additions.
- Efficient handling of signed numbers as negative partial products are managed uniformly.
- Commonly used in high-speed digital arithmetic units such as in processors and digital signal processors.

#### Example of Partial Product Encoding

##### Triplet Bits Operation on Multiplicand

000 or 111 0

001 or 010  $+1 \times$  multiplicand

011  $+2 \times$  multiplicand

100  $-2 \times$  multiplicand

101 or 110  $-1 \times$  multiplicand

This table determines how each triplet in the multiplier translates into multiplying the multiplicand by these factors.

**Tools Used:** Cadence Virtuoso is a comprehensive electronic design automation (EDA) tool widely used for designing and simulating integrated circuits (ICs), especially analog, mixed-signal, and custom digital circuits. It provides an integrated environment for schematic capture, circuit simulation, and layout design, enabling engineers to create full-custom IC designs efficiently.

## 1) Design and implementation of the Radix 4 Booth Multiplier

### Booth Encoder Block

The encoder which is used in Booth multiplication algorithm is called Booth encoder. It is the

encoder which is used to perform Booth encoding. It performs encoding of 3 input signals.

They are: P2, P1, P0. We will have total 4 Booth encoders. Each of these encoders take inputs

from the multiplier bus, X [0:7]. The number of input bits is 3. The encoder actually takes the 3

inputs as a group. These groupings we have mentioned earlier. The output of the encoder also

consists of 3 bits. The outputs are: S, D, N.

S: If Y is multiplied with + or – 1, S will be 1. Otherwise 0.

D: If Y is multiplied with + or – 2, S will be 1. Otherwise 0.

N: If Y is multiplied with -1 or -2 then N will be 1. If multiplied with 1 or 2 then N will be 0

$$S = P1 \text{ Xor } P0$$

$$D = P2 \cdot \overline{P1} + P2 \cdot \overline{P0}$$

$$N = \overline{P2} \cdot P1 \cdot P0 + P2 \cdot \overline{P1} \cdot \overline{P0}$$

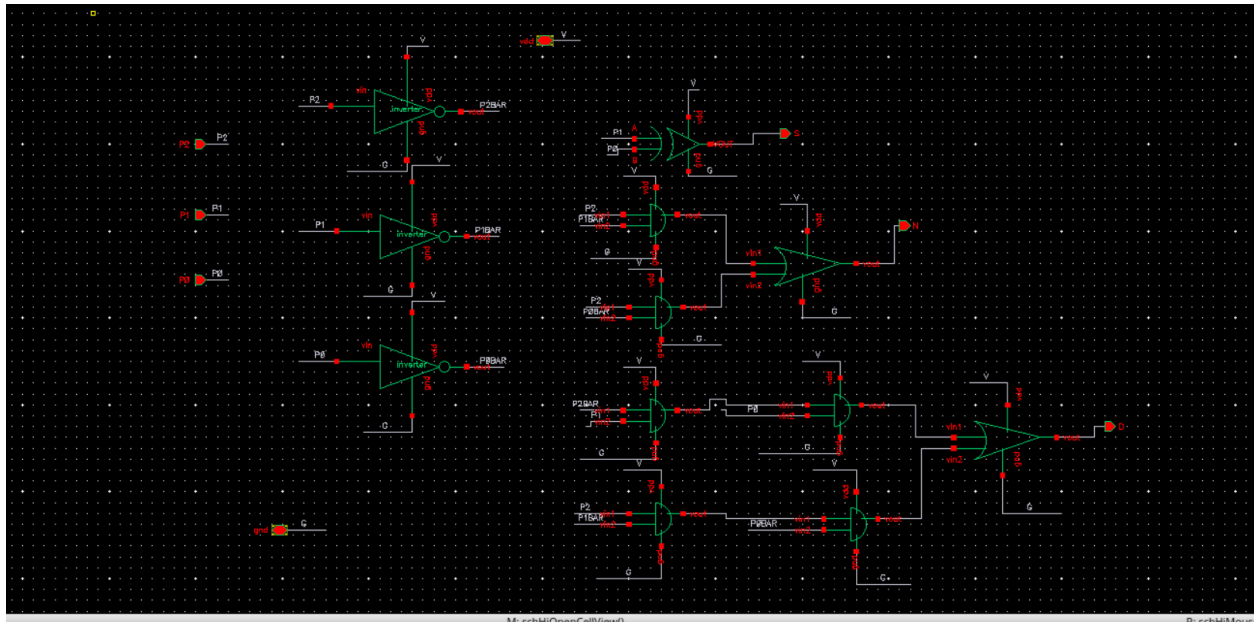


Figure 1.1 Booth Encoder

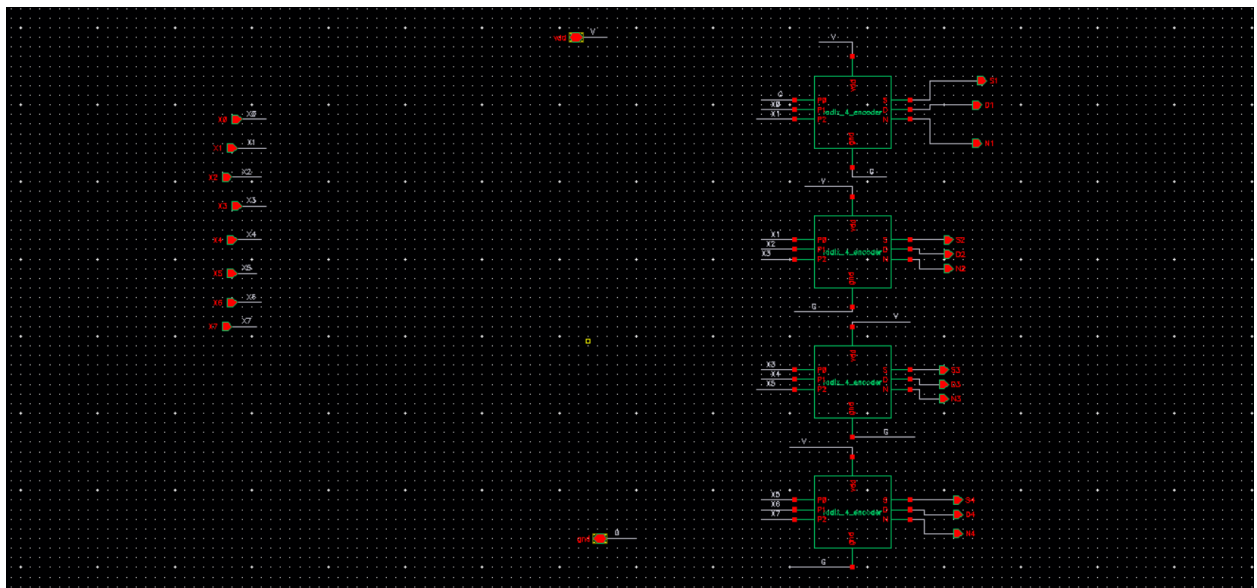


Figure 1.2 8bit Booth Encoder

## Booth Decoder Block

This decoder is driven by the outputs of the Booth Encoder blocks. The multiplicand

Y is distributed to all the rows. The select lines of the booth encoders, S, D and N control the

operation of Booth Decoders. This decoder block works in bit by bit operation.

Actually we need to multiply Y by 1 or 2. If Y is multiplied by 1 then Y remains as it is. If we want to multiply by 2 then one is shifted 1 bit and the LSB becomes 0.

When S=1, it gives the present value of Y as output. Multiplied by 1.

When D=1, it gives the previous value of Y as output. Multiplied by 2.

Let's assume a0, a1 are the data bits from Y.

a0 = represents bit of Y

a1 = represents bit of 2Y

Output will depend on value of S, D.

Output= S.a0 +D.a1

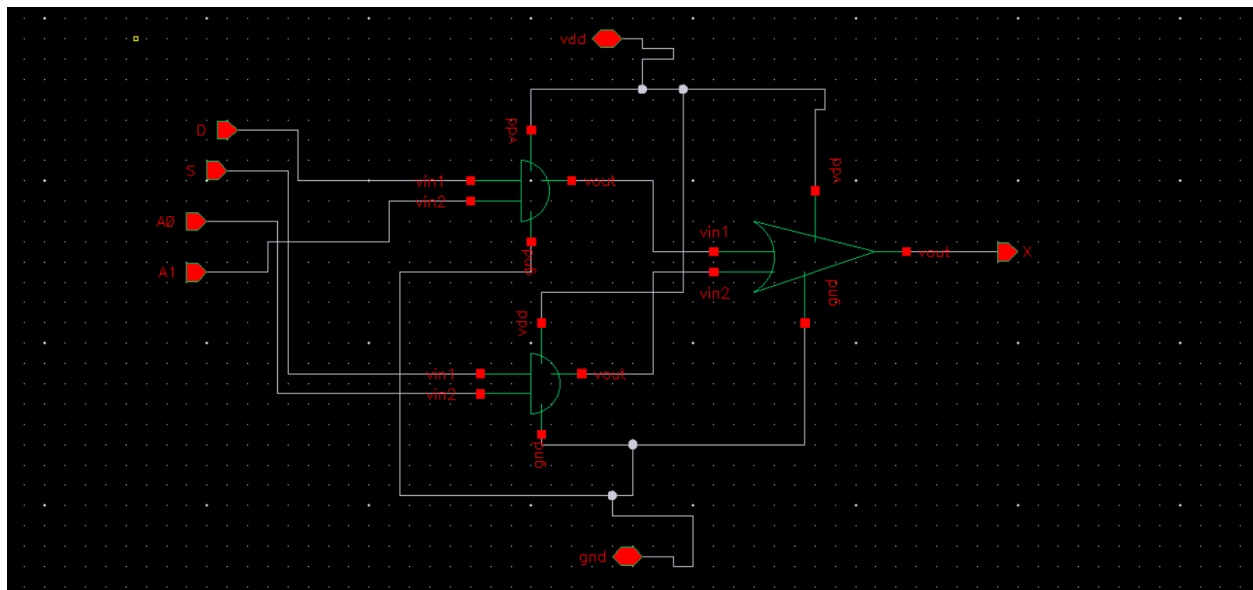


Figure 1.3 Booth Decoder

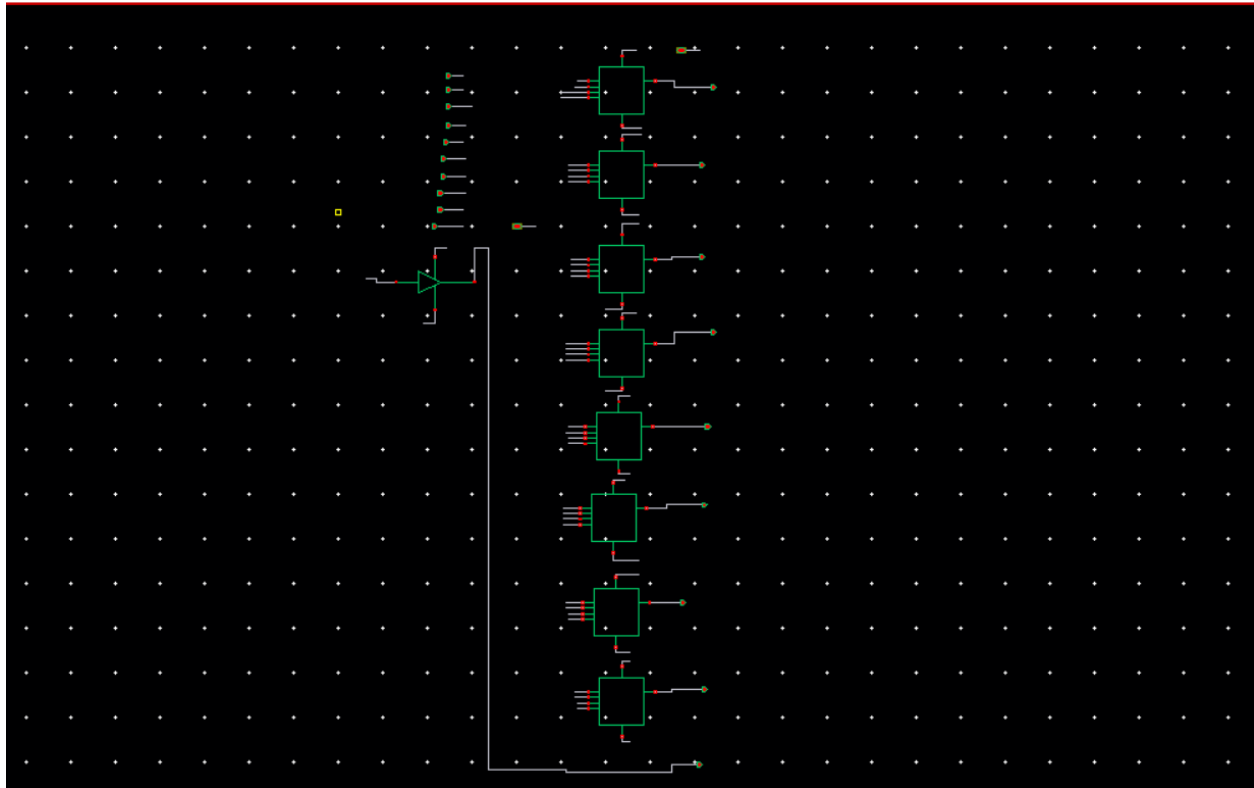


Figure 1.4 8bit Booth Decoder

### 1's complement:

To do 1's complement those 9 bits from previous blocks were xored with 1.

$D \oplus N$ .  $D$  is Data and  $N$  is output of encoder detects whether the bit is signed or unsigned. When  $N=1$  it is treated as negative number when 0 it is positive.

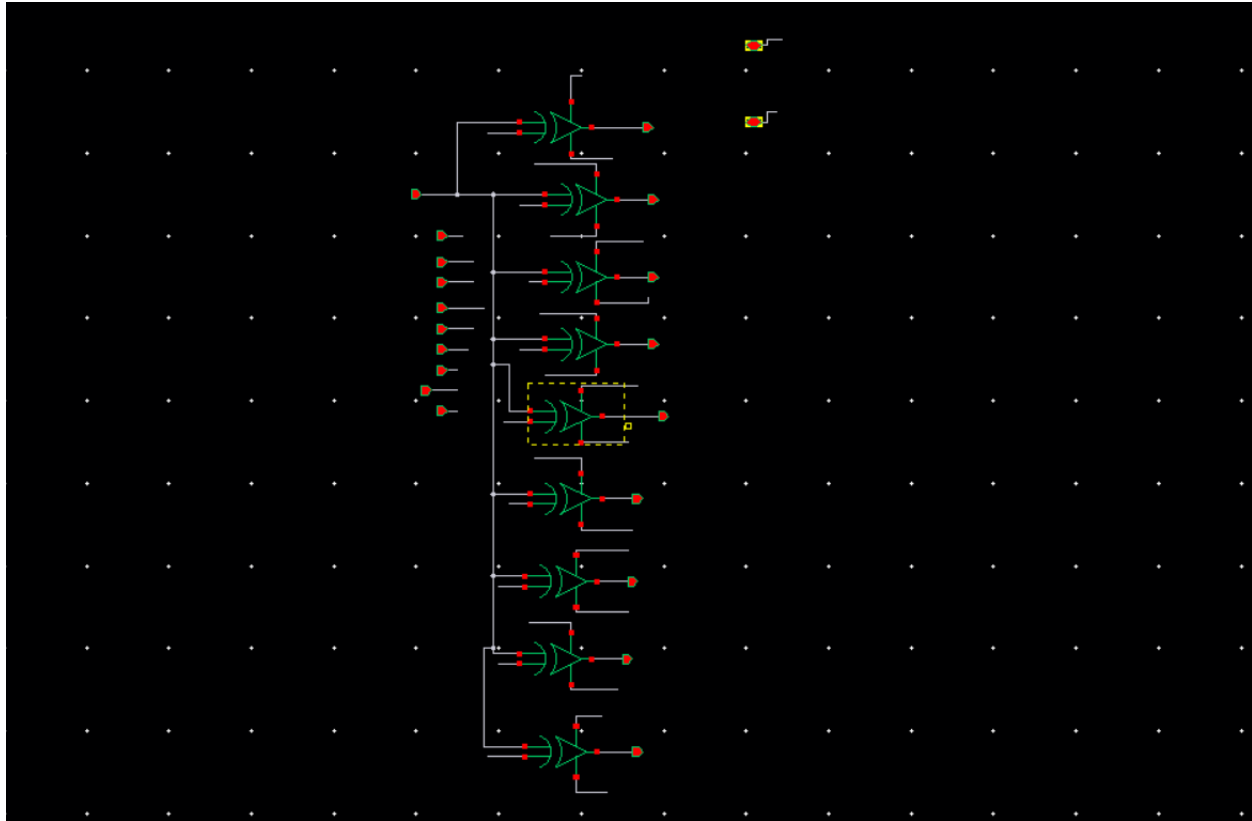


Figure 1.5 1s\_Complement

### 2's complement:

Then that result is added in a 9 bit adder where one input is the output of previous circuit and

another input is 00000000Nb. the result is the 2's complement of Y or  $2Y$ . The circuit diagram

and description of 9 bit adder will be given in kogge stone adder description section.

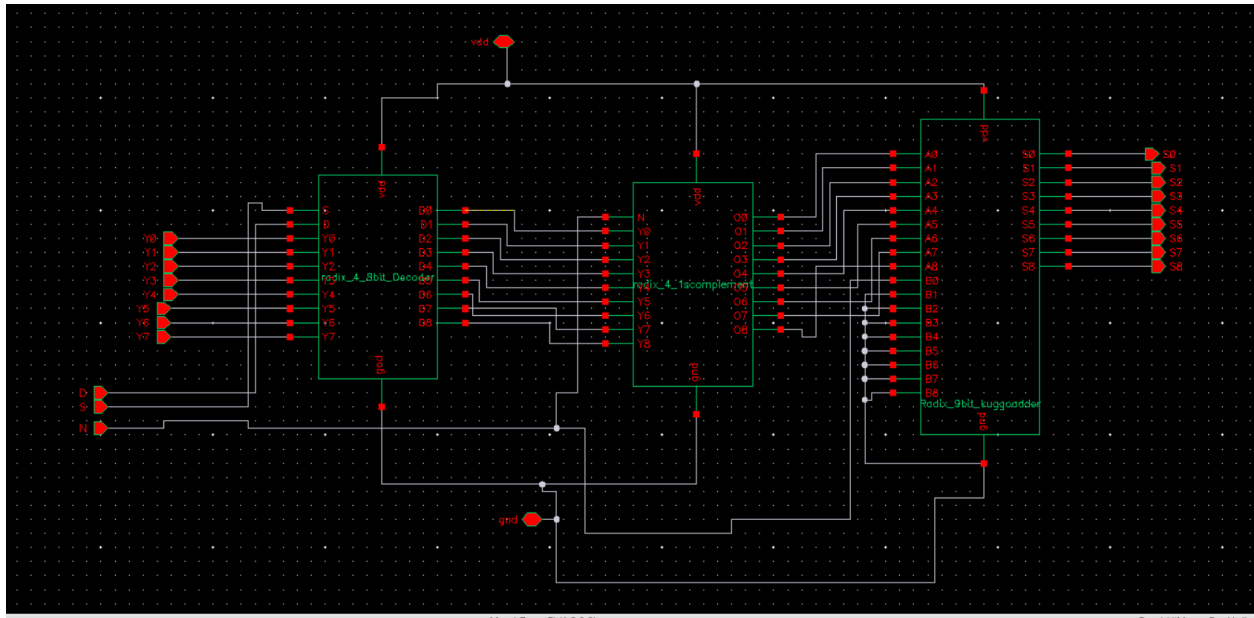


Figure 1.6 Booth Decoder with 2s complement

## Combined Decoder and Encoder

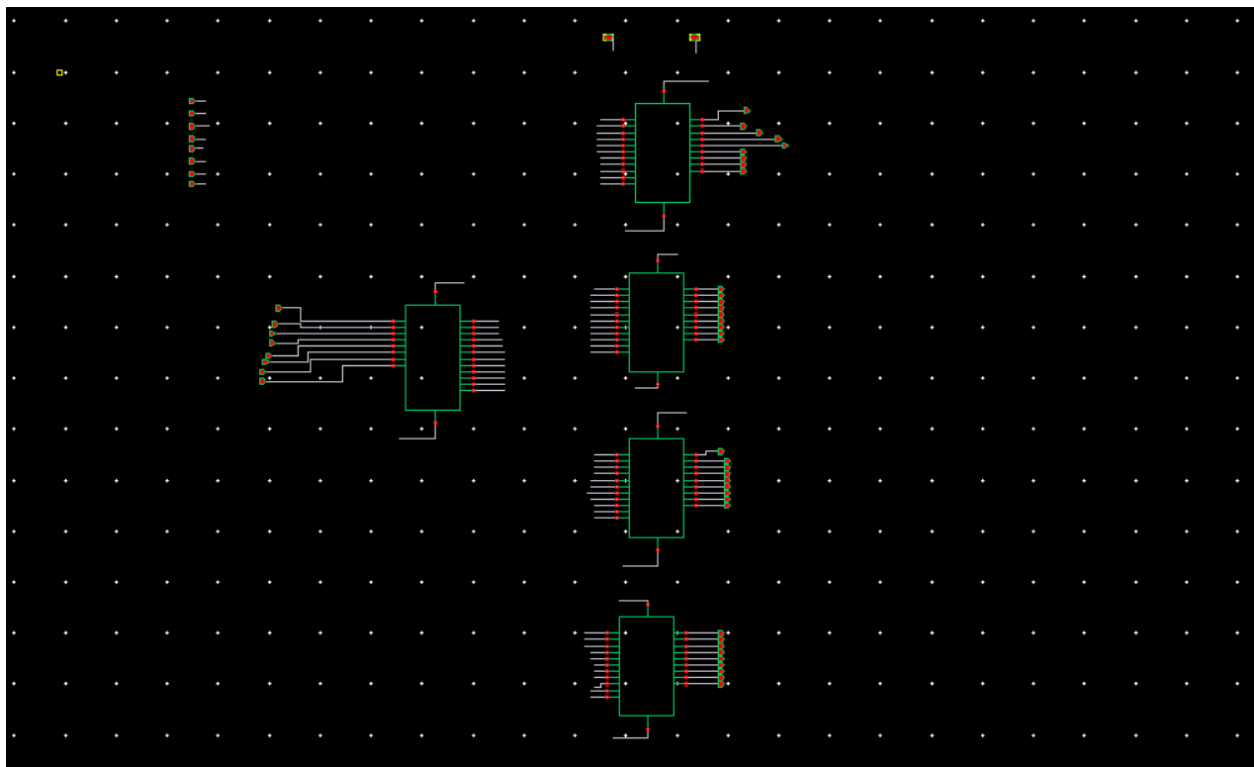


Figure 1.7 Combined Decoder Encoder

## Ripple Carry Adder

A 16-bit Ripple Carry Adder (RCA) is a basic digital circuit used to perform binary addition on two 16-bit numbers. It consists of sixteen cascaded 1-bit full adders connected in series, where the carry output of each full adder is fed as the carry input to the next significant bit's adder. This cascading gives rise to the name "ripple carry," as the carry bits ripple through the adder chain from the least significant bit (LSB) to the most significant bit (MSB).

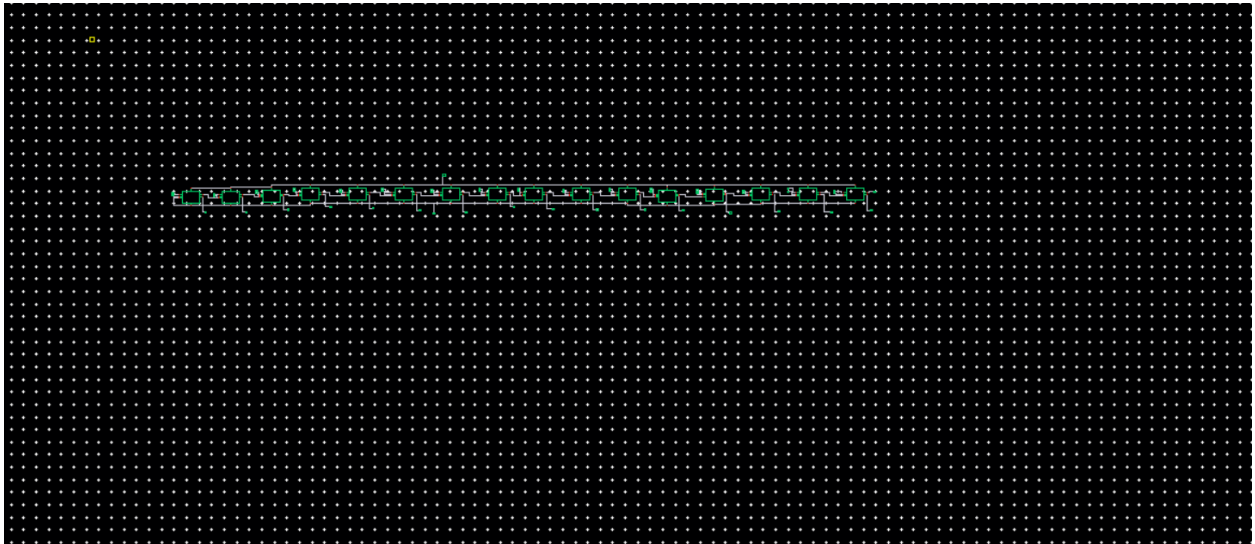


Figure 1.8 Ripple Carry Adder

Multiplier

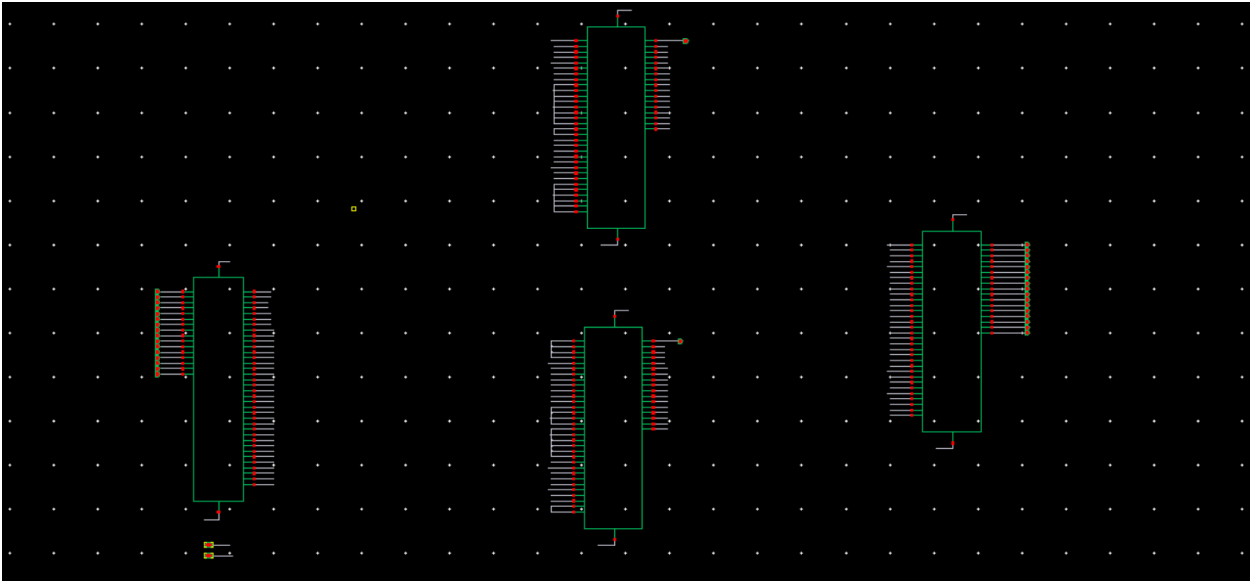


Figure 1.9 Radix\_4\_Booth\_Multiplier

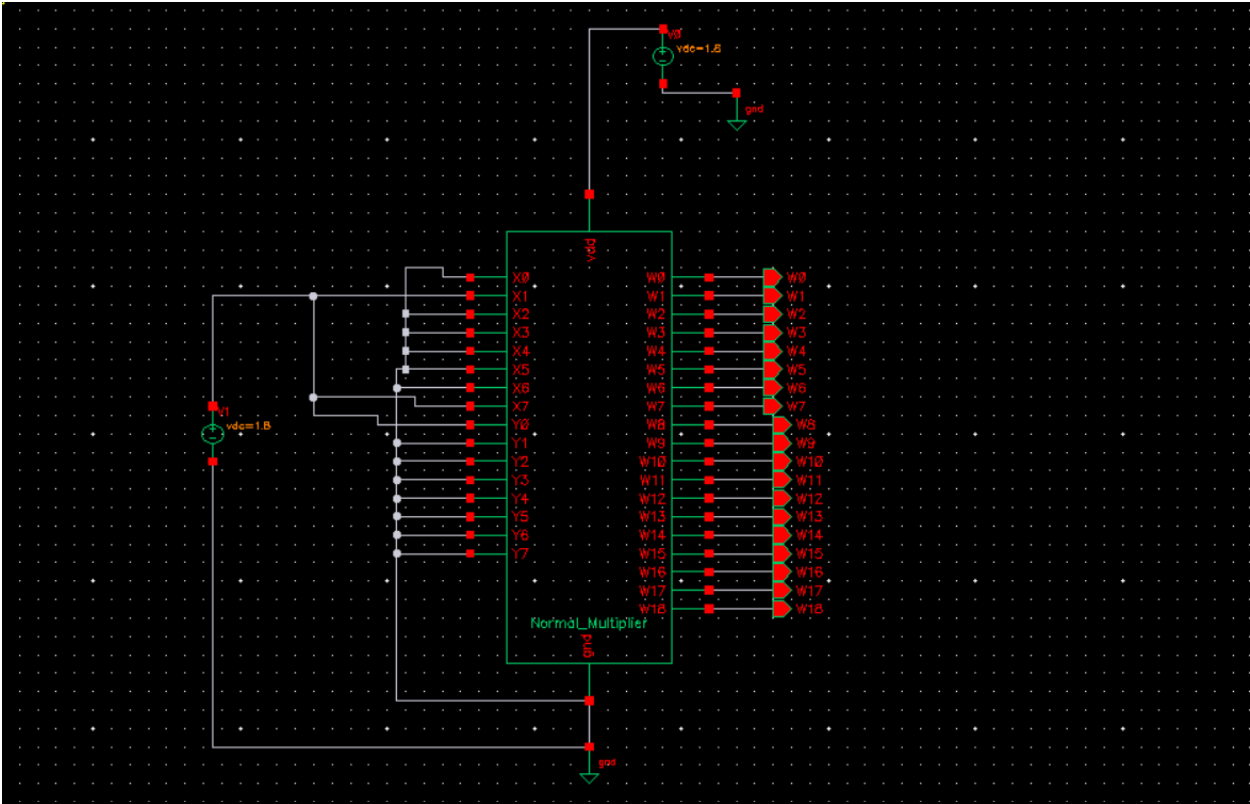


Figure 1.10 Radix\_4\_Booth\_Multiplier

## Test Input

X=1000 0010, Y=0000 0001

W=X.Y=1111 1111 0000 0010

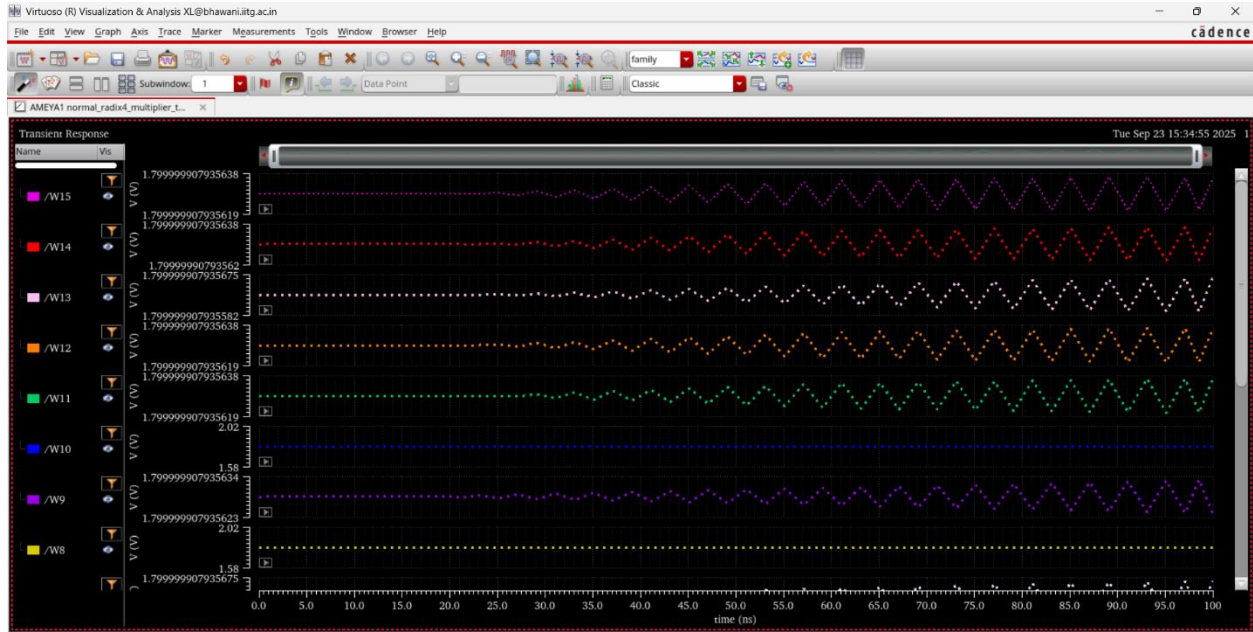


Figure 1.11 Output from MSB to 8<sup>th</sup>

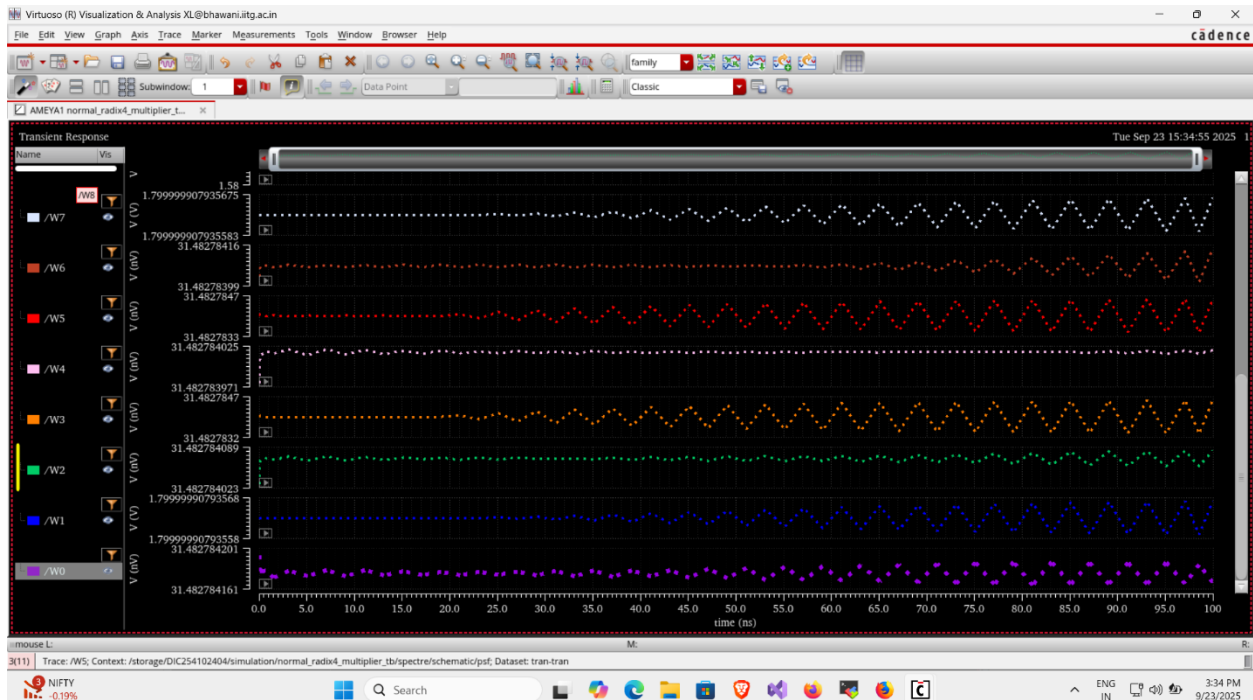


Figure 1.12 Output from 7<sup>th</sup> to LSB

Power=1.527mw

Number Of Transistors Used=Around 2400 Transistors

Input Voltage Time Period=20ns

$T_{\text{delay}}=2\text{ns}$

$T_{\text{fall}}=1\text{ns}$

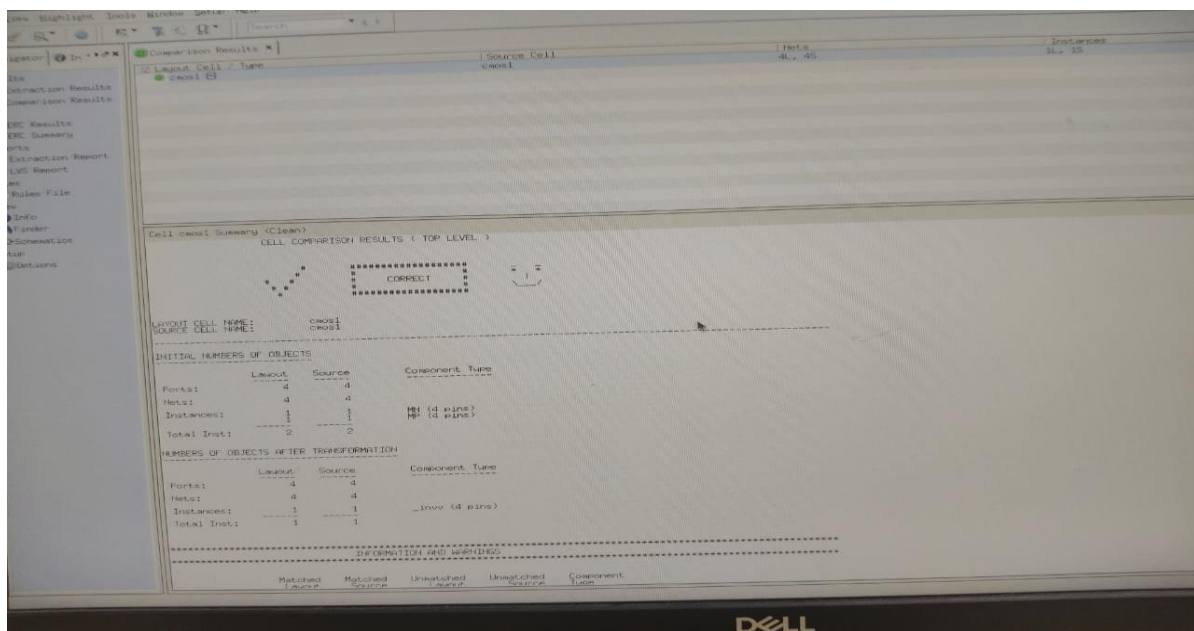
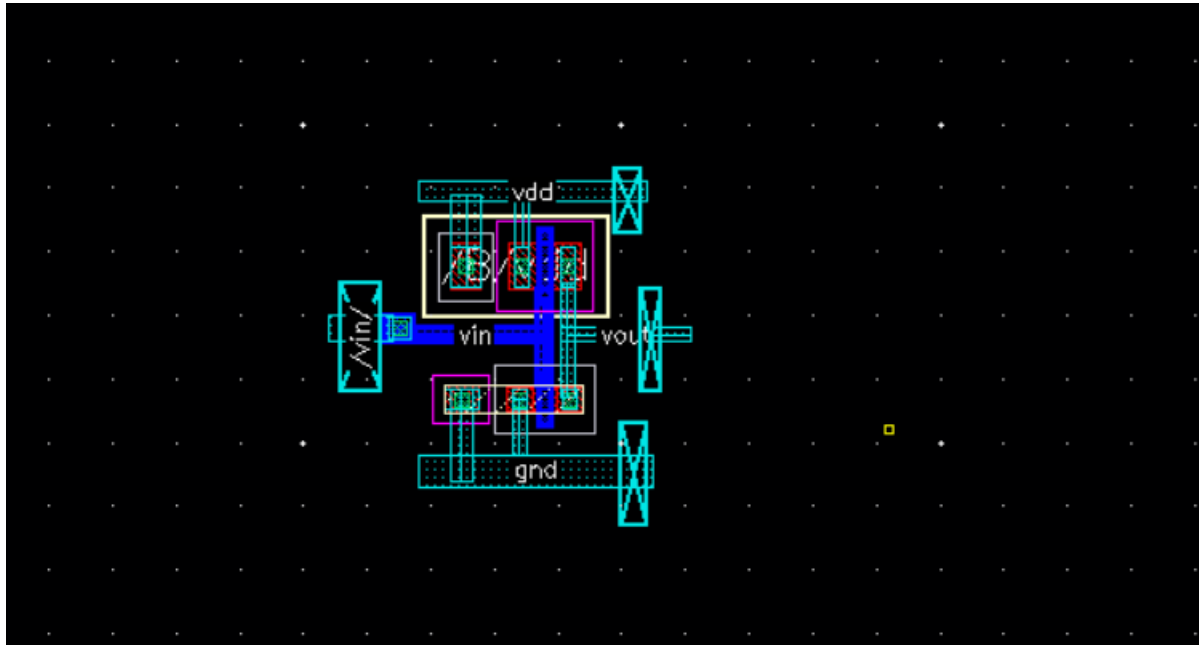
$T_{\text{rise}}=1\text{ns}$

$T_{\text{on}}=10\text{ns}$

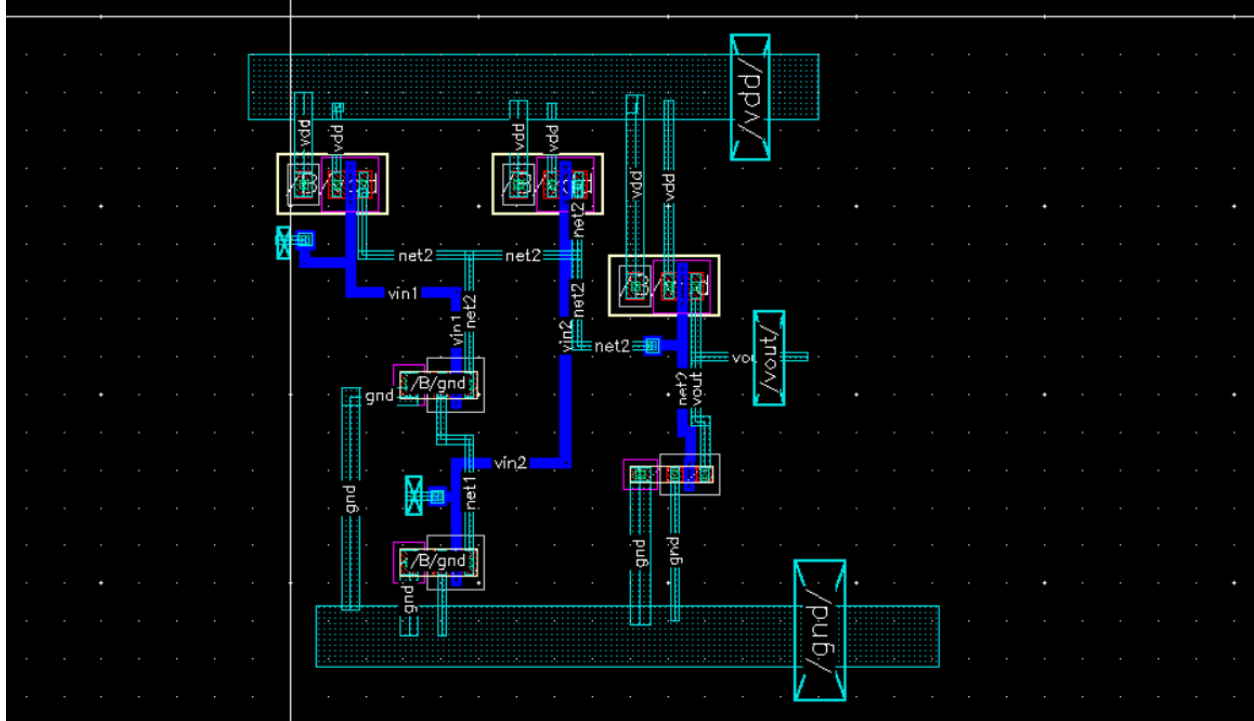
Delay=1.684 ns

# Layout

## Inverter



And



Results  
on Results

Layout Cell / Type  
and\_gate

Source Cell  
and\_gate

Nets  
6L, 6S

Instances  
2L, 2S

Cell and\_gate Summary (Clean)

CELL COMPARISON RESULTS ( TOP LEVEL )

LAYOUT CELL NAME: and\_gate  
SOURCE CELL NAME: and\_gate

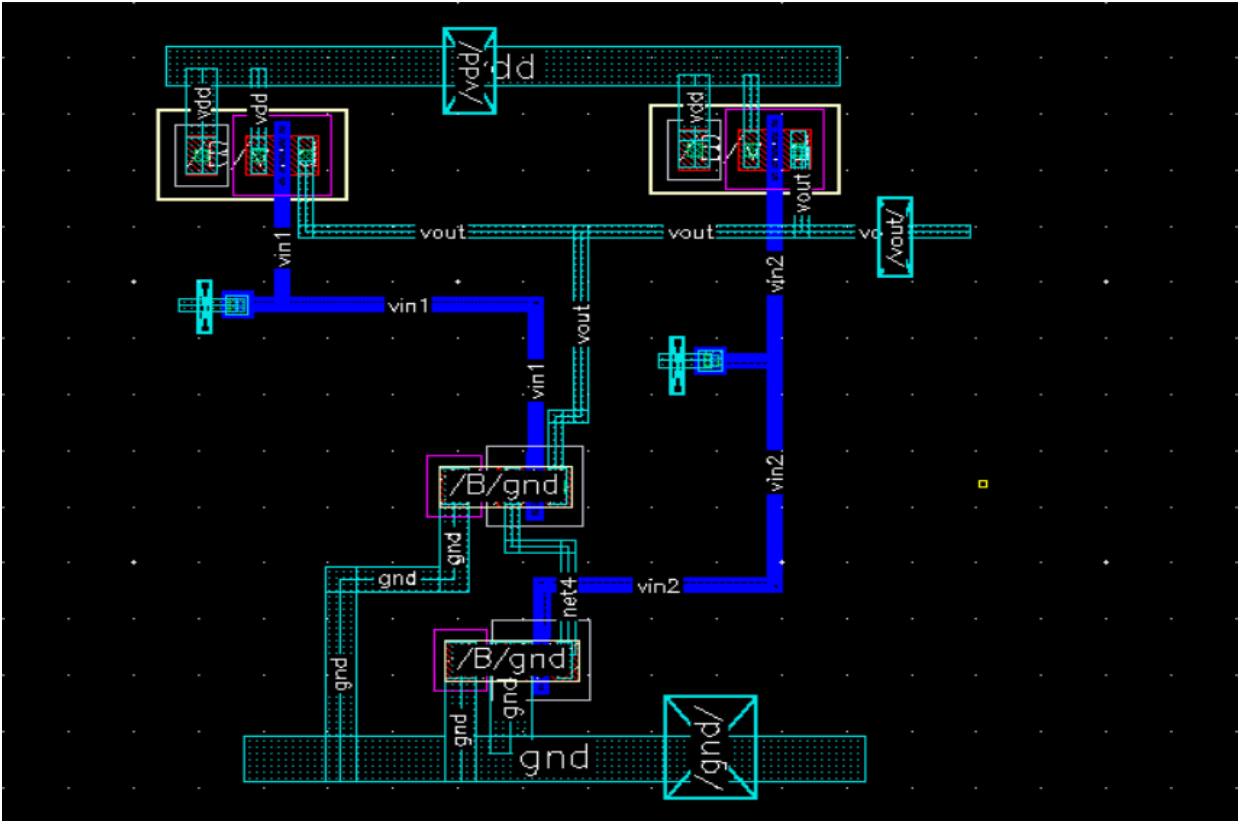
INITIAL NUMBERS OF OBJECTS

|             | Layout | Source | Component Type             |
|-------------|--------|--------|----------------------------|
| Ports:      | 5      | 5      |                            |
| Nets:       | 7      | 7      |                            |
| Instances:  | 3      | 3      | M4 (4 pins)<br>MP (4 pins) |
| Total Inst: | 6      | 6      |                            |

NUMBERS OF OBJECTS AFTER TRANSFORMATION

Windows taskbar with search bar and various application icons.

# Nand



FileViewHighlightToolsWindowSetupHelp

Search

Navigator

Info

Comparison Results

Results

Extraction Results

Comparison Results

ERC

ERC Results

ERC Summary

Reports

Extraction Report

LVS Report

Rules

Rules File

View

Info

Finder

Schematics

Setup

Options

Layout Cell / Type

Source Cell

Nets

Instances

Ports

nand\_gate

nand\_gate

SL, SS

1L, 1S

SL, SS

Cell nand\_gate Summary (Clean)

CELL COMPARISON RESULTS ( TOP LEVEL )

CORRECT

LAYOUT CELL NAME: nand\_gate

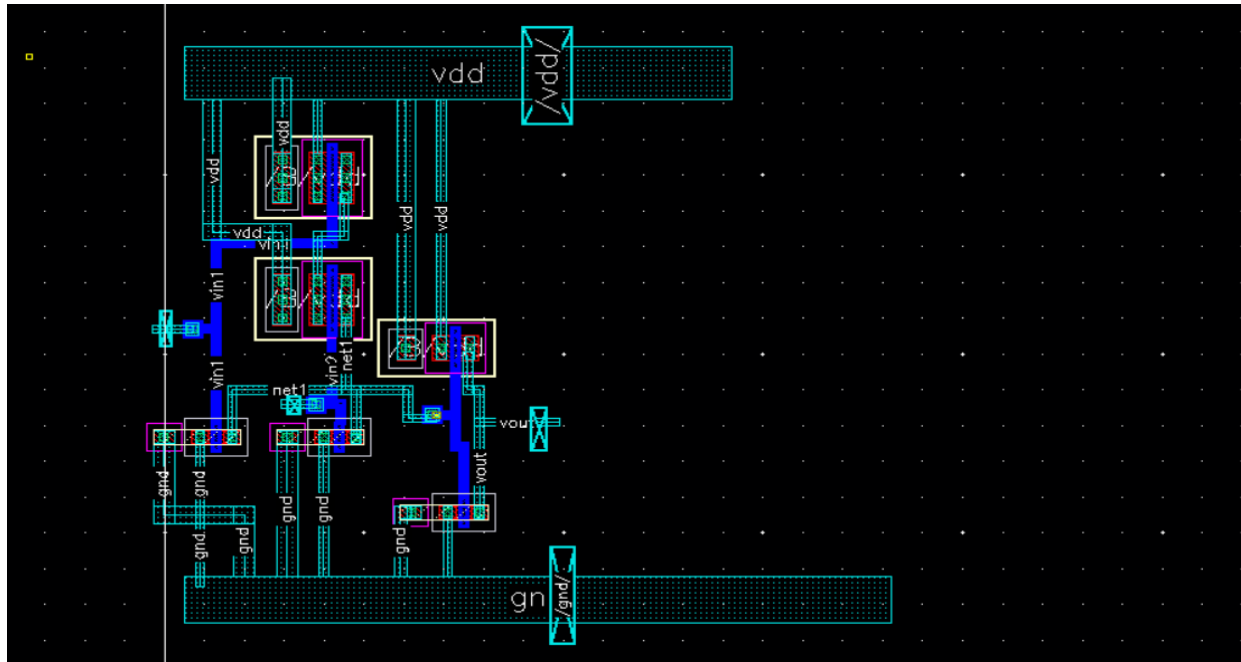
SOURCE CELL NAME: nand\_gate

INITIAL NUMBERS OF OBJECTS

|             | Layout | Source | Component Type |
|-------------|--------|--------|----------------|
| Ports:      | 5      | 5      |                |
| Nets:       | 6      | 6      |                |
| Instances:  | 2      | 2      | MN (4 pins)    |
|             |        |        | MP (4 pins)    |
| Total Inst: | 4      | 4      |                |

NUMBERS OF OBJECTS AFTER TRANSFORMATION

Or



Cell or\_gate Summary (Clean)

CELL COMPARISON RESULTS ( TOP LEVEL )

\*\*\*\*\*  
# CORRECT #  
\*\*\*\*\*

LAYOUT CELL NAME: or\_gate  
SOURCE CELL NAME: or\_gate

-----

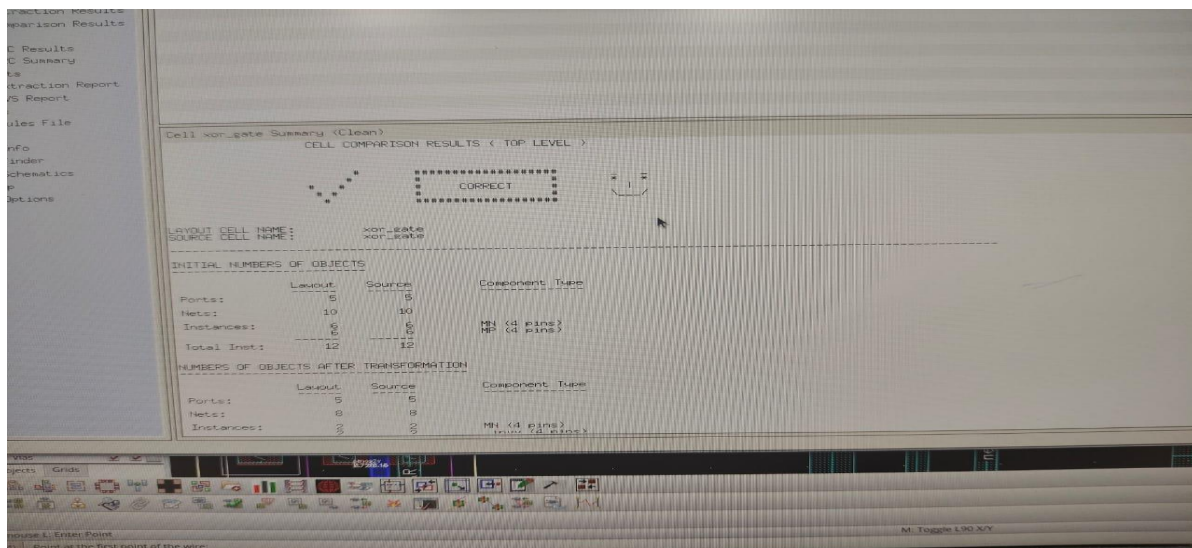
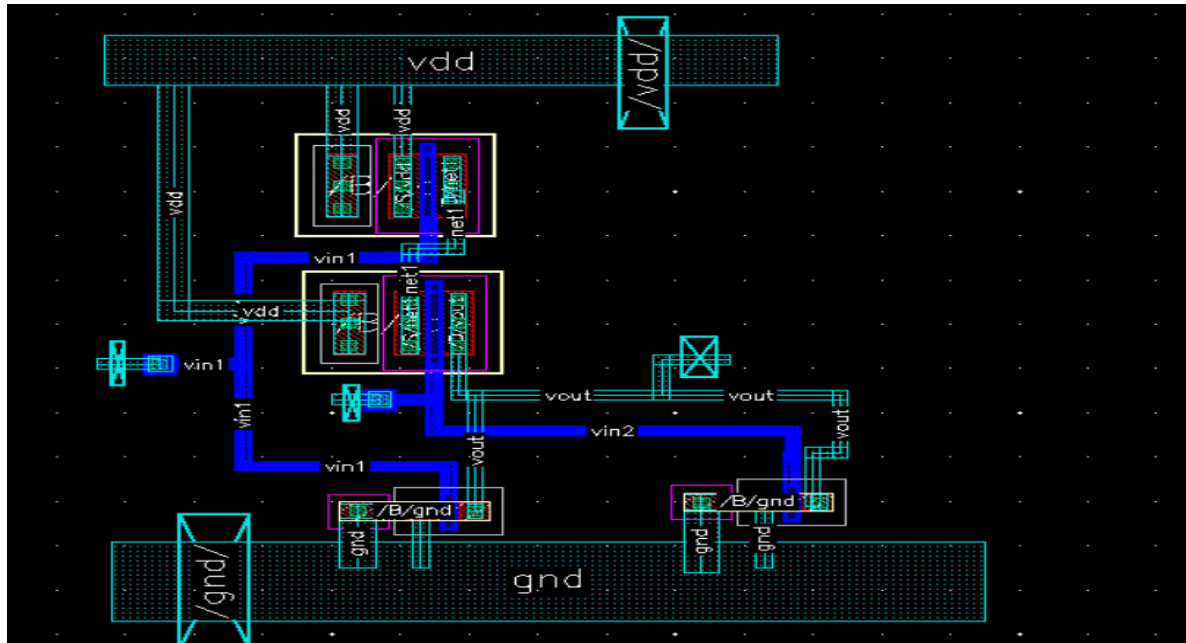
INITIAL NUMBERS OF OBJECTS

|             | Layout | Source | Component Type |
|-------------|--------|--------|----------------|
| Ports:      | 5      | 5      |                |
| Nets:       | 8      | 8      |                |
| Instances:  | 3      | 3      | MN (4 pins)    |
|             | 3      | 3      | MP (4 pins)    |
| Total Inst: | 6      | 6      |                |

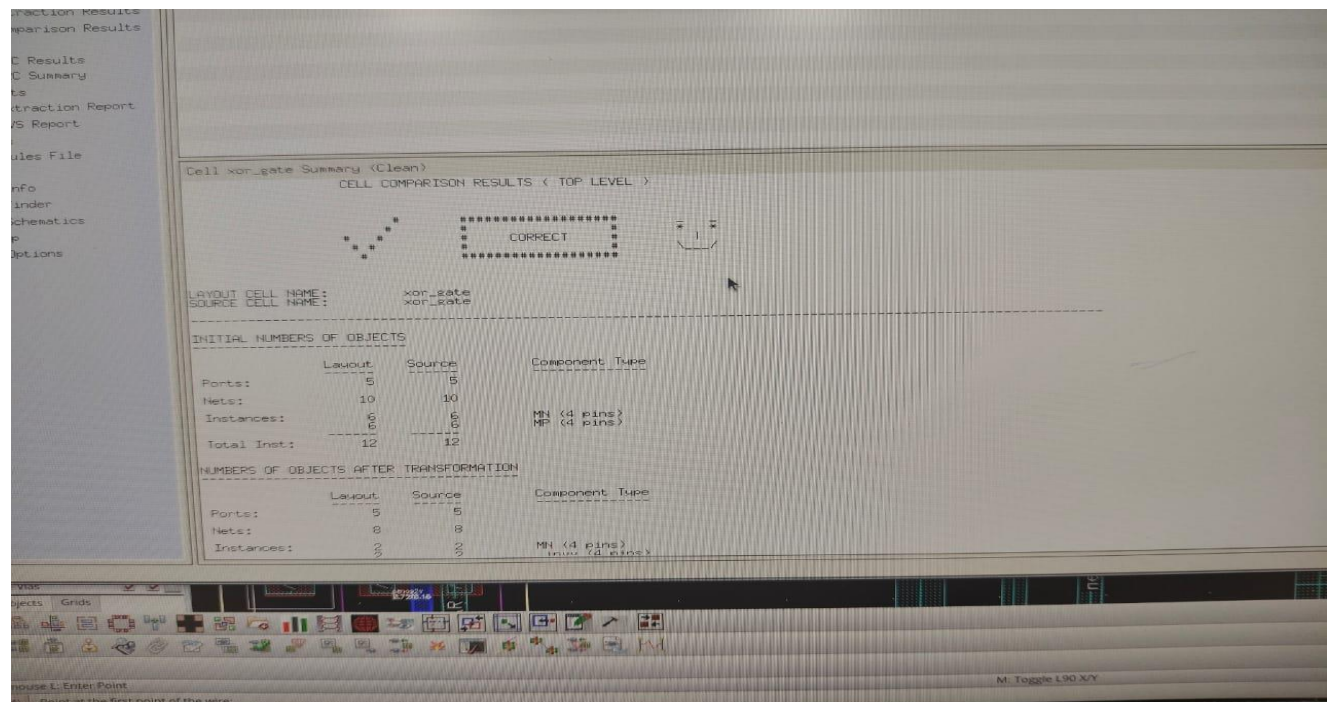
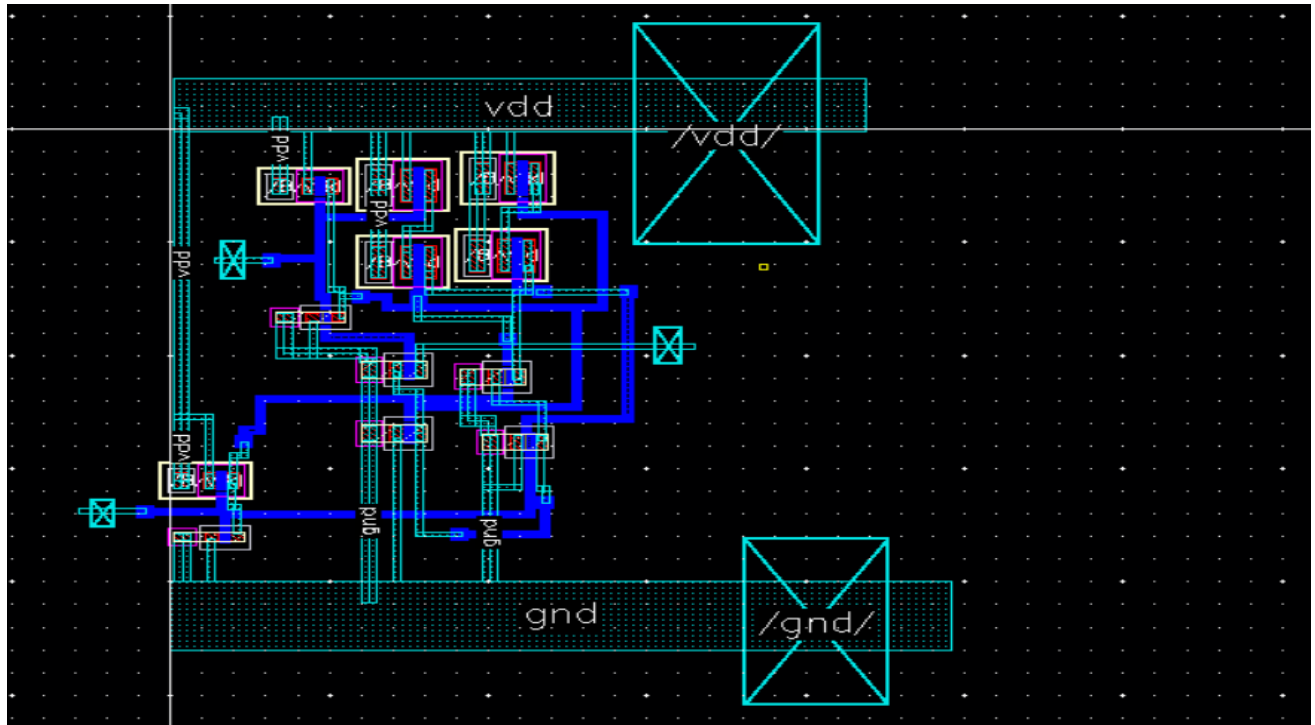
NUMBERS OF OBJECTS AFTER TRANSFORMATION

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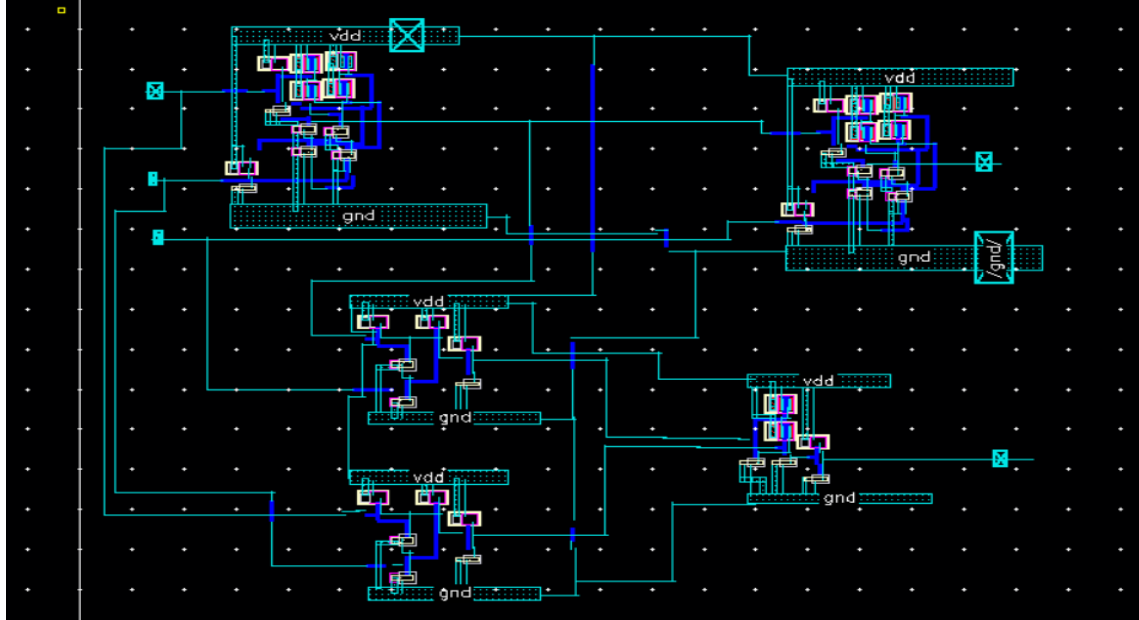
# Nor gate



# Xor



## Full Adder



```
PORT FILE NAME:      full_adder.lvs.report
YOUT NAME:          full_adder.sp ('full_adder')
URCE NAME:          full_adder.src.net ('full_adder')
ILE FILE:           _calibre.lvs_
REATION TIME:       Sun Oct 12 20:41:08 2025
URRENT DIRECTORY:   /storage/DIC254102404
SER NAME:           DIC254102404
ALIBRE VERSION:     v2022.4_37.20      Fri Dec 2 17:43:09 PST 2022
```

## OVERALL COMPARISON RESULTS

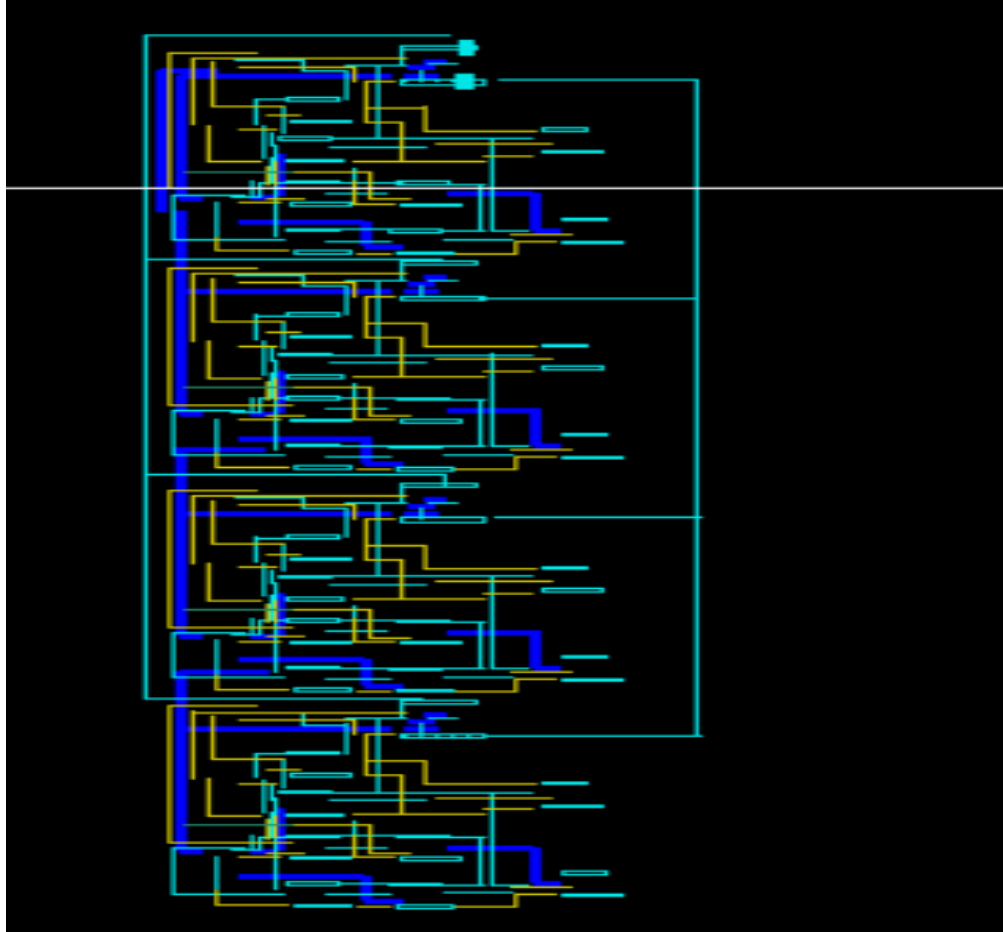
CELL SUMMARY

```

*****
Result      Layout      Source
-----
CORRECT     full_adder

```

# Encoder



Cell radix\_4\_8bit\_encoder Summary (Clean)

CELL COMPARISON RESULTS < TOP LEVEL >

\*\*\*\*\*  
\* CORRECT \*  
\*\*\*\*\*

LAYOUT CELL NAME: radix\_4\_8bit\_encoder  
SOURCE CELL NAME: radix\_4\_8bit\_encoder

---

INITIAL NUMBERS OF OBJECTS

|             | Layout | Source | Component   | Type |
|-------------|--------|--------|-------------|------|
| Ports:      | 22     | 22     |             |      |
| Nets:       | 150    | 150    |             |      |
| Instances:  | 132    | 132    | MN (4 pins) |      |
|             | 132    | 132    | MP (4 pins) |      |
| Total Inst: | 264    | 264    |             |      |

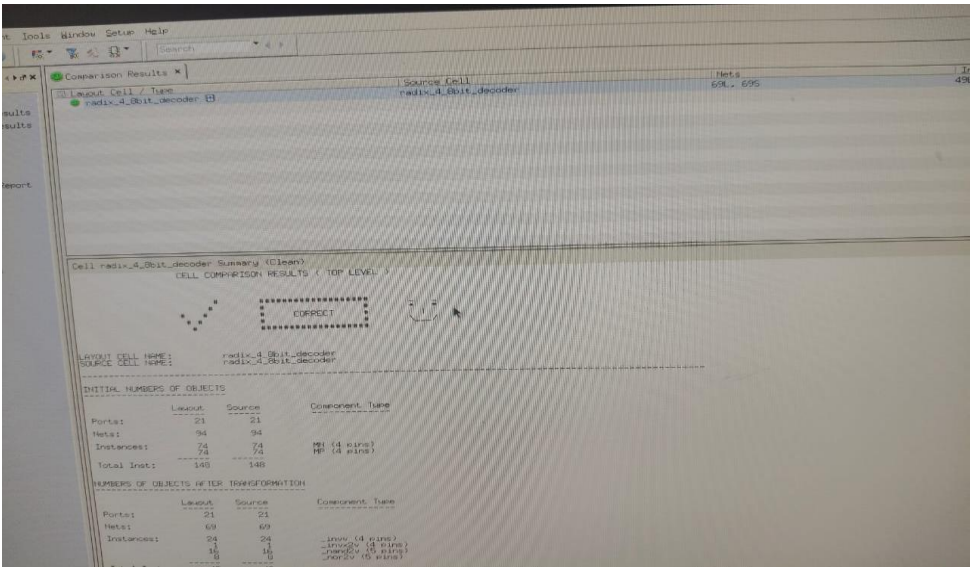
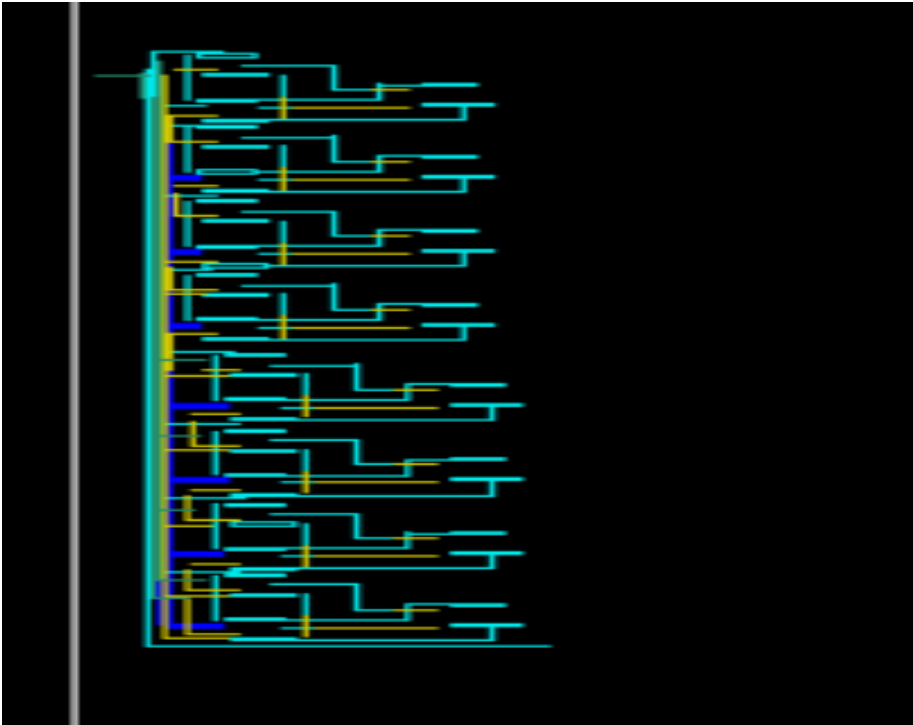
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NUMBERS OF OBJECTS AFTER TRANSFORMATION

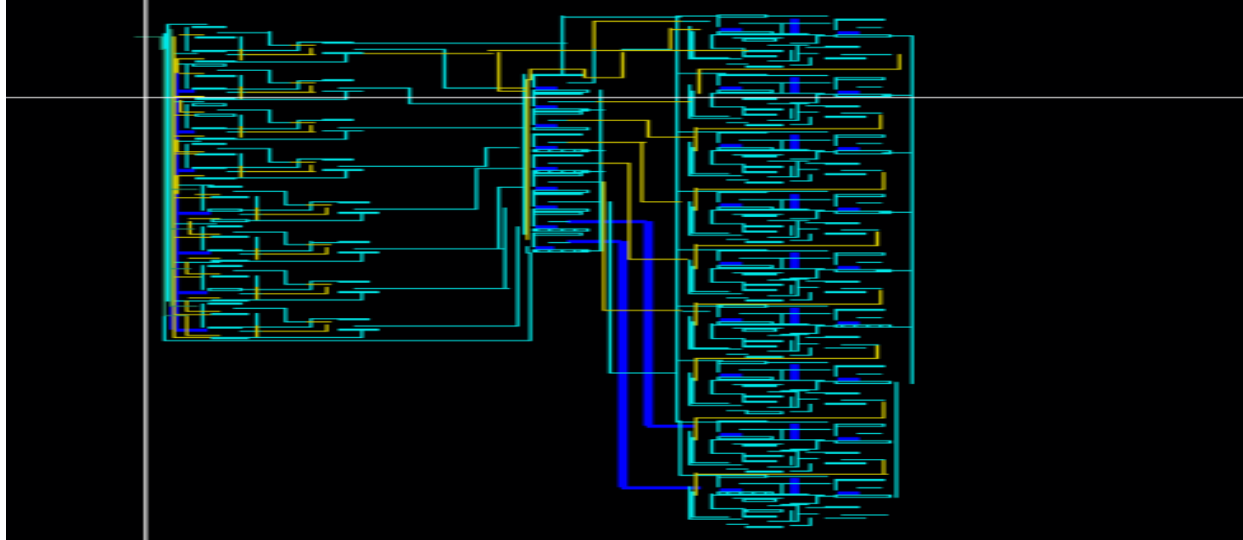
|  | Layout | Source | Component | Type |
|--|--------|--------|-----------|------|
|--|--------|--------|-----------|------|

Type here to search

Decoder



## 2s complement



Cell 2s\_complement Summary (Clean)  
CELL COMPARISON RESULTS ( TOP LEVEL )

\*\*\*\*\*  
CORRECT  
\*\*\*\*\*

LAYOUT CELL NAME: 2s\_complement  
SOURCE CELL NAME: 2s\_complement

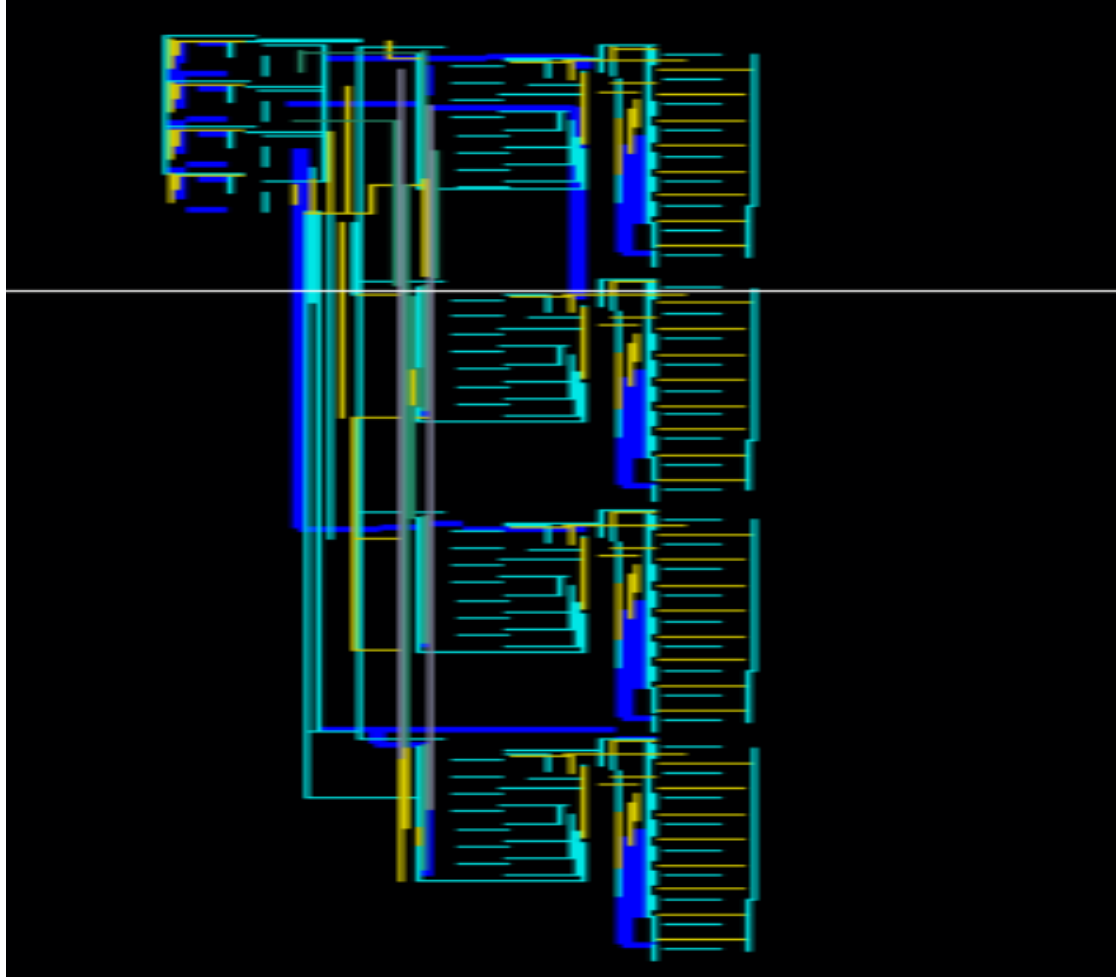
-----  
INITIAL NUMBERS OF OBJECTS  
-----

|             | Layout | Source | Component Type             |
|-------------|--------|--------|----------------------------|
| Ports:      | 23     | 23     |                            |
| Nets:       | 330    | 330    |                            |
| Instances:  | 317    | 317    | MN (4 pins)<br>MP (4 pins) |
| Total Inst: | 634    | 634    |                            |

-----  
NUMBERS OF OBJECTS AFTER TRANSFORMATION  
-----

|            | Layout | Source | Component Type                                                            |
|------------|--------|--------|---------------------------------------------------------------------------|
| Ports:     | 23     | 23     |                                                                           |
| Nets:      | 224    | 224    |                                                                           |
| Instances: | 105    | 105    | _invv (4 pins)<br>_invx2v (4 pins)<br>_nand2v (6 pins)<br>_nor2b (7 pins) |
|            | 34     | 34     |                                                                           |
|            | 27     | 27     |                                                                           |

## Combined Decoder Encoder



Comparison Results

Results Summary

Comparison Report

Report

File

Statistics

Summary

Cell combined\_decoder\_encoder Summary (Clean)

CELL COMPARISON RESULTS ( TOP LEVEL )

\*\*\*\*\*  
CORRECT  
\*\*\*\*\*

LAYOUT CELL NAME: combined\_decoder\_encoder  
SOURCE CELL NAME: combined\_decoder\_encoder

-----

INITIAL NUMBERS OF OBJECTS

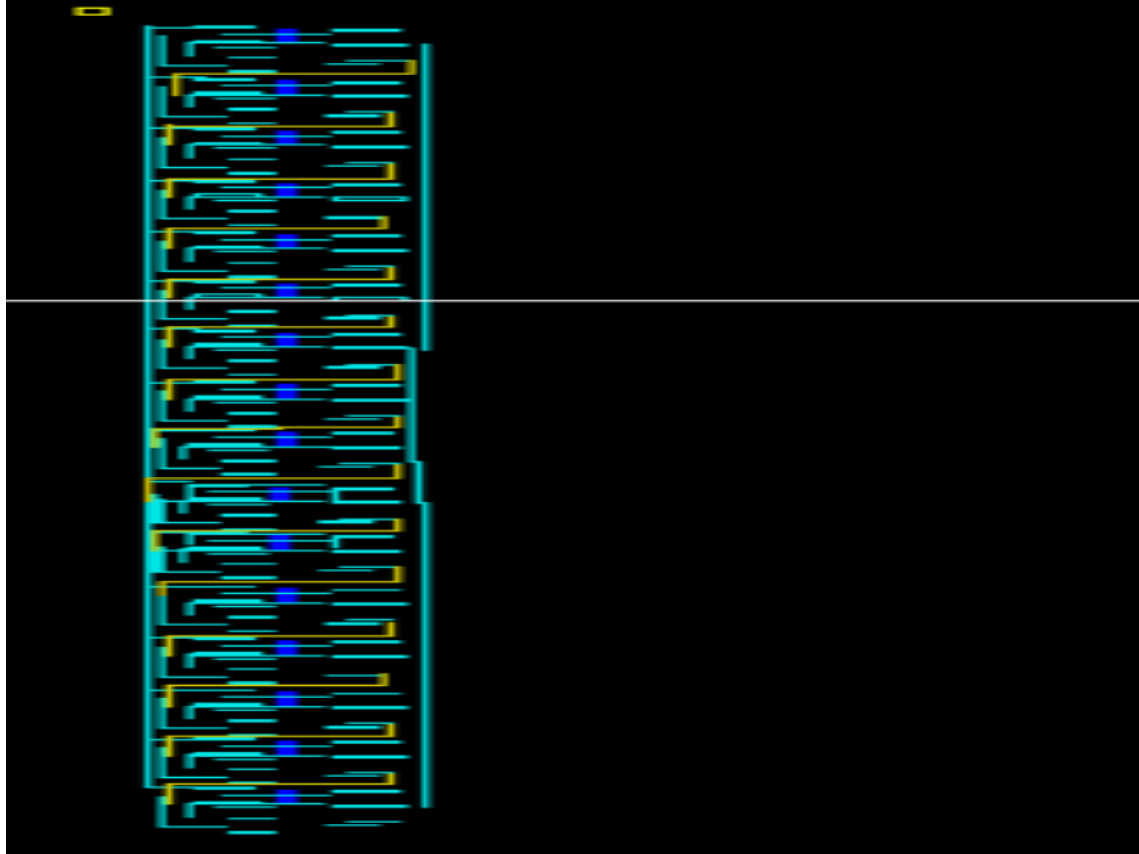
|             | Layout | Source | Component | Type     |
|-------------|--------|--------|-----------|----------|
| Ports:      | 58     | 58     |           |          |
| Nets:       | 1418   | 1418   |           |          |
| Instances:  | 1400   | 1400   | MP        | (4 pins) |
|             | 1400   | 1400   | MP        | (4 pins) |
| Total Inst: | 2800   | 2800   |           |          |

-----

NUMBERS OF OBJECTS AFTER TRANSFORMATION

-----

## 16 bit rca



Comparison Results

| Layout Cell / Type     | Source Cell        | Nets       | Instan     |
|------------------------|--------------------|------------|------------|
| ripple_carry_adder (S) | ripple_carry_adder | 2581, 2585 | 2581, 2585 |

Cell ripple\_carry\_adder Summary (Clean)

CELL COMPARISON RESULTS ( TOP LEVEL )

\*\*\*\*\*  
\* CORRECT \*  
\*\*\*\*\*

LAYOUT CELL NAME: ripple\_carry\_adder  
SOURCE CELL NAME: ripple\_carry\_adder

-----

INITIAL NUMBERS OF OBJECTS

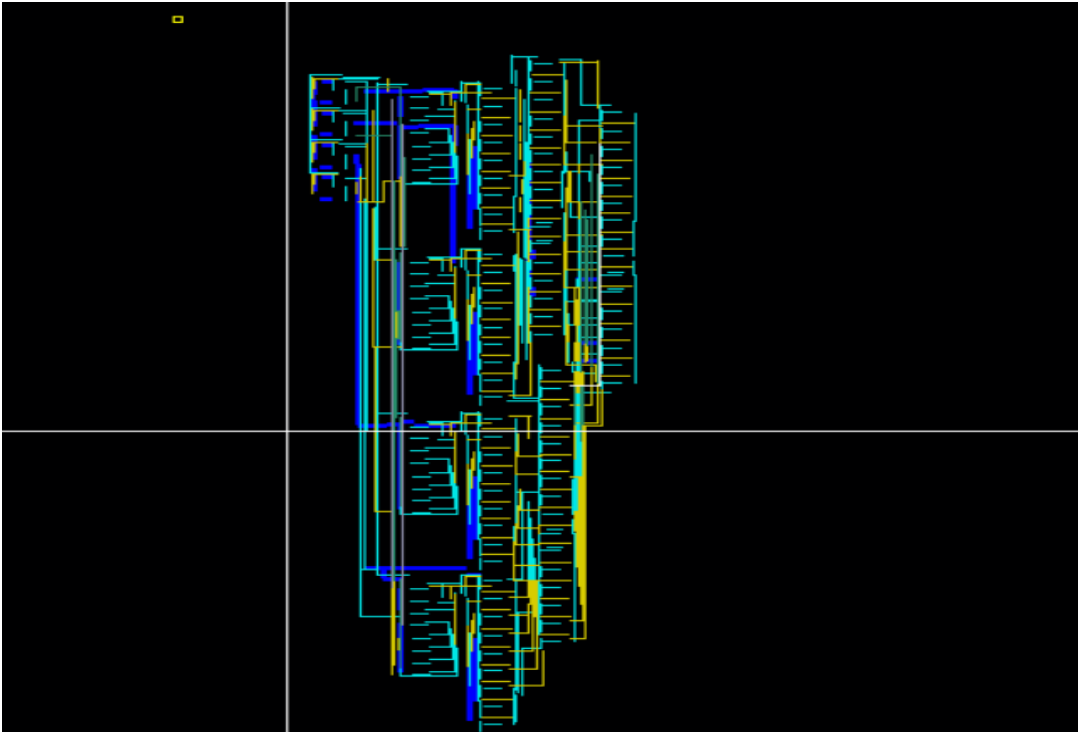
|             | Layout | Source | Component Type             |
|-------------|--------|--------|----------------------------|
| Ports:      | 51     | 51     |                            |
| Nets:       | 370    | 370    |                            |
| Instances:  | 336    | 336    | MN (4 pins)<br>MP (4 pins) |
| Total Inst: | 672    | 672    |                            |

-----

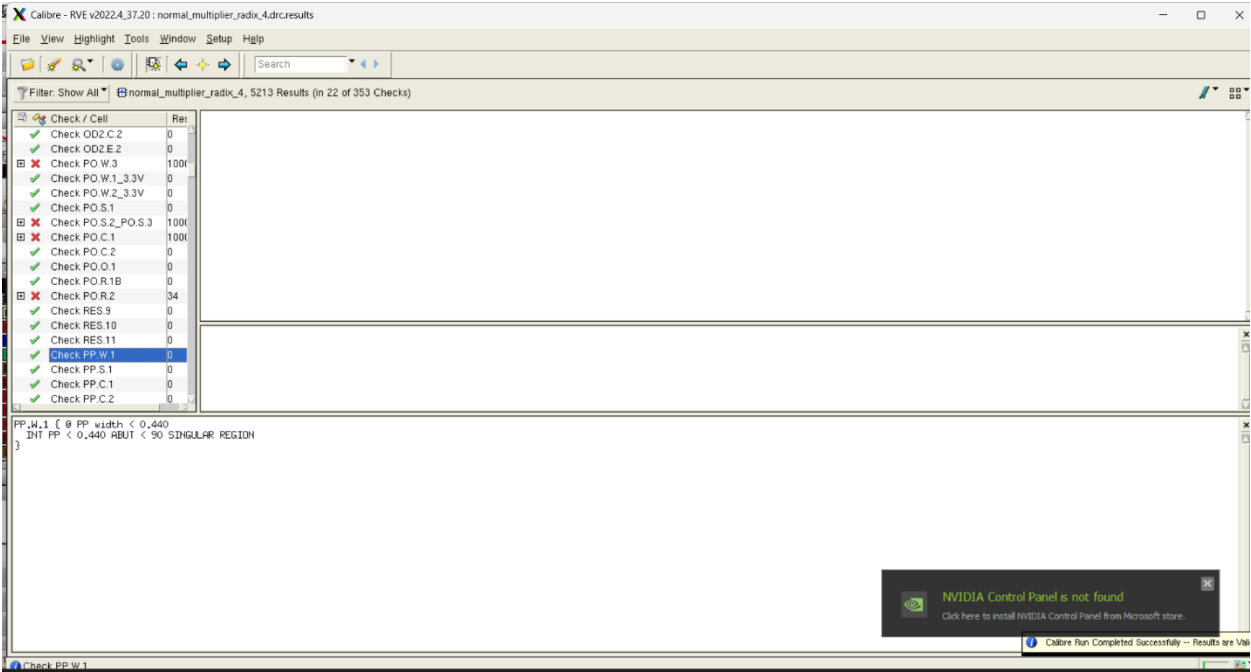
NUMBERS OF OBJECTS AFTER TRANSFORMATION

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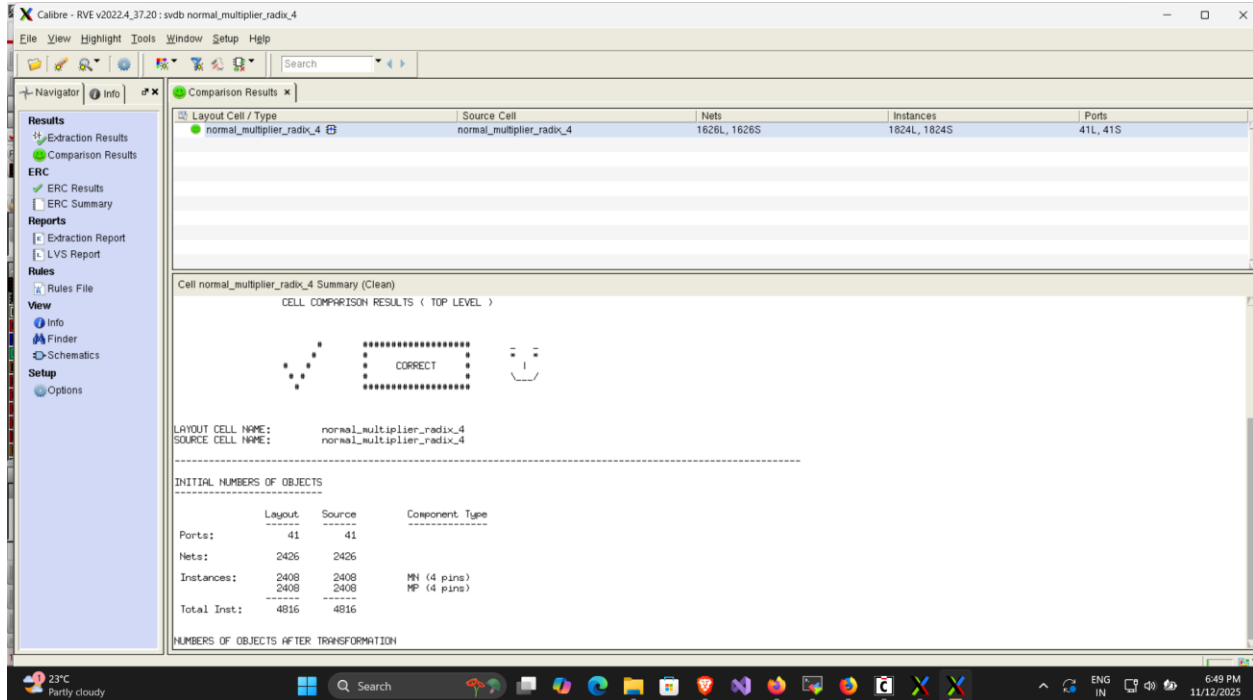
# Multiplier



# DRC



# LVS



Number of Transistors: Around 2400 Transistors

Area: 2.89 mm sq

## Result And Discussion

The layout of the 8-bit Radix-4 Booth Multiplier was successfully designed and verified. DRC and LVS checks were clean.